Scott Phillip Ahrenkiel

Applied Physics - AI Data Trainer

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Professional Summary

Ph.D. Physicist with 30+ years of experience in theoretical and computational physics, currently training large language models on complex physics problems. Expert in classical mechanics, electrodynamics, statistical mechanics, quantum mechanics, and materials physics with exceptional analytical writing demonstrated through 100+ peer-reviewed publications and a published textbook. Proven ability to formulate rigorous physics problems, develop detailed solutions with meticulous attention to units and mathematical proof structure, and identify errors in physical reasoning.

Relevant Experience

AI Trainer

Mercor (Remote) | *February 2026 - Present*

- Create training data for large language models (LLMs) by formulating, testing, and refining complex physics problems at PhD and research level
- Design problems spanning classical mechanics, electrodynamics, statistical mechanics, and quantum mechanics
- Develop detailed solutions demonstrating rigorous mathematical reasoning with meticulous attention to units, scientific notation, and logical proof structure
- Audit AI-generated responses for physical accuracy, identifying violations of fundamental principles (conservation laws, thermodynamics, quantum mechanics)
- Provide structured feedback to enhance model understanding of physics-informed reasoning

Physicist/Programmer

eQuillTech, Arvada, Colorado | *October 2025 - Present*

- Develop physics-based software integrating computational modeling with analytical solutions
- Create algorithms for materials science and physics simulations requiring deep understanding of fundamental physics principles

Computational Scientist

Big Ladder Software, Denver, Colorado | *June 2023 - October 2025*

- Developed physics-based simulation library (HPWHsim) for heat pump water heaters with emphasis on thermodynamic rigor and energy conservation
- Created interactive analysis tools using Python (Plotly Dash, NumPy, SciPy) for parameter refinement and validation
- Maintained 100+ unit and function tests ensuring physical accuracy across all simulations
- Implemented formal data representations (JSON) for external model specifications with rigorous physical constraints

Professor (Full/Associate/Assistant)

South Dakota School of Mines and Technology | *August 2006 - July 2021*

Teaching & Problem Development: - Developed and taught graduate courses in nanocharacterization, crystallography, nanoelectronics, and photovoltaics requiring mastery of quantum mechanics, electrodynamics, and statistical mechanics - Created comprehensive problem sets and instructional experiments demonstrating physical principles across multiple domains - Formulated exam problems testing deep understanding of fundamental physics at graduate level - Advised 15+ PhD students on complex physics problems in materials science and nanoscience

Research & Computational Physics: - Developed theoretical models for transmission electron microscopy contrast requiring quantum mechanical treatment of electron-matter interactions - Created dynamical electron-diffraction software for computational analysis using numerical methods - Applied convergent-beam electron diffraction to extract alloy compositions and strain states from first principles - Implemented automation software for electron microscopy with rigorous calibration and error analysis

Senior Scientist I / Scientist II / Postdoctoral Scientist

National Renewable Energy Laboratory (NREL) | *January 1995 - July 2006*

- Developed physical models of strain relaxation in metamorphic semiconductor heterostructures for world-record efficiency photovoltaic devices
- Created theoretical models for TEM image contrast of partially ordered structures based on dynamical electron diffraction theory
- Developed electron-diffraction simulation software implementing rigorous quantum mechanical scattering theory

- Applied statistical mechanics and thermodynamics to phase separation and ordering phenomena in semiconductor alloys

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Education

Ph.D., Physics University of Colorado, Boulder | *Completed December 1995* - Dissertation: Phase Separation and Atomic Ordering in Epitaxial Semiconductor Alloys Studied by Transmission Electron Microscopy - Research combined quantum mechanics, statistical mechanics, and computational methods - Developed theoretical models for diffraction contrast and ordering phenomena

M.S., Physics Colorado School of Mines, Golden, Colorado | *Completed December 1991* - Thesis: Intercalation Processes in Layered Silicates and High-Tc Superconducting Oxides Studied by X-Ray Diffraction - Developed complete XRD data acquisition/analysis software package from scratch

B.S., Engineering Physics University of Colorado, Boulder | *Completed December 1987*

Physics Expertise

Classical Mechanics: Lagrangian and Hamiltonian mechanics, continuum mechanics, elasticity theory, strain and stress analysis in crystalline materials

Electrodynamics: Maxwell's equations, electromagnetic wave propagation, electron-photon interactions, optical properties of materials

Quantum Mechanics: Schrödinger equation, perturbation theory, quantum mechanical scattering theory, electron diffraction, electronic structure of solids, density functional theory applications

Statistical Mechanics: Thermodynamics, phase transitions, ordering phenomena, equilibrium and non-equilibrium statistical mechanics, kinetic theory

Specialized Areas: Solid state physics, materials physics, semiconductor physics, crystal structure and symmetry, diffraction theory, computational physics

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Technical Skills

Programming & Scientific Computing: - Python (NumPy, SciPy, Matplotlib) - extensive use for physics simulations and data analysis - MATLAB - data analysis, numerical methods, visualization - C++ - simulation software development, algorithm implementation - LabVIEW, Igor Pro - experiment automation and data acquisition - Digital Micrograph scripting - microscopy automation

Computational Methods: - Numerical solution of differential equations - Monte Carlo methods - Dynamical electron diffraction simulation - Finite element analysis - Statistical data analysis - Algorithm development for physics problems

Data Analysis & Validation: - Rigorous error analysis and uncertainty quantification - Unit consistency verification across complex calculations - Scientific notation and dimensional analysis - Validation against fundamental physical principles - Statistical analysis of experimental and computational data

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Analytical Writing & Communication

Published Textbook: - Co-author, *Theory and Methods of Photovoltaic Material Characterization*, World Scientific, 2019 - Comprehensive treatment of semiconductor physics, quantum mechanics of photovoltaic devices, and characterization methods

Book Chapter: - "Diffraction and imaging of ordered semiconductors" in *Spontaneous Ordering in Semiconductor Alloys*, Kluwer Academic, 2002 - Detailed theoretical treatment of electron diffraction and quantum mechanical image formation

Peer-Reviewed Publications: 100+ publications in premier physics and materials science journals demonstrating: - Clear explanation of complex physical phenomena - Rigorous mathematical derivations - Meticulous attention to units and notation - Logical proof structure from first principles - Critical analysis of experimental and theoretical results

Selected Publications Demonstrating Physics Rigor: - "Optimization of buffer layers for lattice-mismatched epitaxy…" *Solar Energy Materials and Solar Cells* (2007) - thermodynamic analysis of strain relaxation - "Nanoparticle shape and configuration analysis by transmission electron tomography" *Journal of Microscopy* (2008) - 3D reconstruction from quantum mechanical diffraction - "Solid-phase crystallization kinetics…" *Materials Science and Engineering B* (2011) - statistical mechanics of phase transitions - "Standards-based method for compositional analysis by energy dispersive x-ray spectrometry…" *Microscopy and Microanalysis* (2013) - rigorous quantitative analysis with multivariate statistics

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Teaching Experience

Developed and taught graduate-level physics courses requiring creation of complex problems and detailed solutions:

- **NANO 703** - Instrumentation for Characterization of Nanomaterials (quantum mechanics, electrodynamics)

- **NANO 704** - Crystallography and Structure of Nanomaterials (group theory, symmetry operations)
- **NANO 705** - Nanoelectronic Materials (quantum mechanics, solid state physics, statistical mechanics)
- **NANO 708** - Nanomaterials for Photovoltaics (electrodynamics, semiconductor physics, thermodynamics)
- **Selected Topics Courses** - Advanced topics in strain relaxation, ordered semiconductors, defects in nanomaterials

Created comprehensive problem sets, examinations, and instructional experiments testing deep understanding of fundamental physics principles with meticulous attention to mathematical rigor and unit consistency.

Patents & Inventions

- **US20220271194A1** - Strain balanced direct bandgap aluminum indium phosphide quantum wells for light emitting diodes (quantum mechanics, band structure engineering)
- **US20060048700A1** - Method for achieving device-quality, lattice-mismatched, heteroepitaxial active layers (thermodynamics, kinetics)

Professional Certifications

- Machine Learning (Stanford/Coursera, 2024)
- Supervised Machine Learning (Stanford/Coursera, 2024)
- Unsupervised Learning (Stanford/Coursera, 2024)

Professional Memberships

- American Physical Society
- Microscopy Society of America
- Materials Research Society

Additional Qualifications

- **Summer Student, CERN** (1987) - Installed time-projection chamber for SPS collider experiment, developed test programs for data acquisition electronics
- **Grants Awarded:** $3.4M+ in competitive research grants from DOE and NSF
- **Graduate Student Supervision:** Advised 15+ PhD students on complex physics problems in quantum mechanics, statistical mechanics, and

electrodynamics
- **Peer Review:** Extensive experience reviewing physics manuscripts for rigorous accuracy and logical consistency

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Why I'm Ideal for This Role

I am currently performing exactly this role at Mercor, creating physics training data for LLMs. My unique combination of:

1. **Active AI training experience** - Currently formulating and refining complex physics problems for LLM training
2. **Deep physics expertise** - PhD-level mastery across all required domains (classical mechanics, electrodynamics, statistical mechanics, quantum mechanics)
3. **Exceptional writing** - 100+ publications and textbook demonstrating clear explanation of complex physics
4. **Meticulous rigor** - Career-long practice of unit consistency, dimensional analysis, and logical proof structure
5. **Programming proficiency** - Extensive Python (NumPy/SciPy) and MATLAB experience
6. **Problem design experience** - 15 years creating graduate-level physics problems and solutions as professor

...makes me uniquely qualified to design physics problems, create rigorous solutions, and audit AI responses for physical accuracy.